

Cepstral Analysis & MFCCs

BSAI – 7th Semester

Assignment # 2

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December 2025

Objective

To understand numerical cepstral analysis and Mel-Frequency Cepstral Coefficients (MFCCs) through step-by-step computation of real cepstrum, complex cepstrum, pitch estimation, Mel scale conversion, and MFCC calculation.

Question 1

Given magnitude spectrum:

$$|X[k]| = [8, 4, 2, 4]$$

Step 1: Compute $\ln |X[k]|$

$$\ln(8) = 2.079, \quad \ln(4) = 1.386, \quad \ln(2) = 0.693$$

$$\ln |X[k]| = [2.079, 1.386, 0.693, 1.386]$$

Step 2: Real Cepstrum

$$c[n] = \frac{1}{4} \sum_{k=0}^3 \ln |X[k]| e^{j\frac{2\pi}{4}kn}$$

For $n = 0$:

$$c[0] = \frac{1}{4}(2.079 + 1.386 + 0.693 + 1.386) = \boxed{1.386}$$

For $n = 1$:

$$c[1] = \frac{1}{4}(2.079 - 0.693) = \boxed{0.346}$$

Question 2

Given STFT spectrum:

$$S[k] = [6, 2j, 1, -2j]$$

(a) Magnitude and Log Magnitude

$$|S[k]| = [6, 2, 1, 2]$$

$$\ln |S[k]| = [1.792, 0.693, 0, 0.693]$$

(b) Phase

$$\angle S[k] = [0, \frac{\pi}{2}, 0, -\frac{\pi}{2}]$$

(Already unwrapped)

(c) Complex Cepstrum

$$\ln S[k] = \ln |S[k]| + j\angle S[k]$$

$$c[n] = \text{IFFT}(\ln S[k])$$

First four coefficients obtained numerically.

Question 3

Given:

$$n = 25, \quad f_s = 10\,000 \text{ Hz}$$

$$F_0 = \frac{f_s}{n} = \frac{10000}{25}$$

$$\boxed{F_0 = 400 \text{ Hz}}$$

Question 4

Given:

$$|X[k]| = [3, 1, 0.5, 1]$$

Log Magnitude

$$\ln |X[k]| = [1.099, 0, -0.693, 0]$$

Cepstrum

$$c[n] = \frac{1}{4} \sum_{k=0}^3 \ln |X[k]| e^{j\frac{2\pi}{4}kn}$$

$$c[0] = \frac{1}{4}(1.099 - 0.693) = \boxed{0.1015}$$

$$c[1] = \frac{1}{4}(1.099 + 0.693) = \boxed{0.448}$$

Question 5

Given:

$$S[k] = [4 - j, 2, 1 + j, 2]$$

Magnitude

$$|S[k]| = [\sqrt{17}, 2, \sqrt{2}, 2]$$

Log Magnitude

$$\ln |S[k]| = [1.417, 0.693, 0.347, 0.693]$$

Cepstrum $c[0]$

$$c[0] = \frac{1}{4}(1.417 + 0.693 + 0.347 + 0.693) = \boxed{0.788}$$

Question 6

Mel scale formula:

$$m = 2595 \log_{10} \left(1 + \frac{f}{700} \right)$$

$$f = 500 \Rightarrow m = \boxed{607 \text{ Mel}}$$

$$f = 1500 \Rightarrow m = \boxed{1357 \text{ Mel}}$$

$$f = 3500 \Rightarrow m = \boxed{2146 \text{ Mel}}$$

Question 7

Given:

$$E = [12, 20, 10, 5, 2]$$

Log Energies

$$\ln(E) = [2.485, 3.0, 2.303, 1.609, 0.693]$$

First MFCC

$$c[1] = \sum_{k=1}^5 \ln(E_k) \cos \left[\frac{\pi}{5}(k - 0.5) \right]$$

$$\boxed{c[1] \approx 2.31}$$

Question 8

Given:

$$n = 40, \quad f_s = 8000$$

$$F_0 = \frac{8000}{40} \Rightarrow \boxed{200 \text{ Hz}}$$

Question 9

Given poles:

$$0.8e^{\pm j\pi/4}$$

Cepstral formula:

$$c[n] = \frac{2(0.8)^n \cos(n\pi/4)}{n}$$

$$c[1] = \boxed{1.131}, \quad c[2] = \boxed{0.32}, \quad c[3] = \boxed{-0.241}$$

Question 10

Given:

$$L = [2.3, 3.0, 2.5, 1.9, 1.2, 0.8]$$

$$c[2] = \sum_{k=1}^6 L_k \cos \left[\frac{2\pi}{6} (k - 0.5) \right]$$

$$\boxed{c[2] \approx -1.84}$$

Conclusion

This assignment provided a strong numerical understanding of cepstral analysis and MFCCs. Step-by-step computations demonstrated pitch estimation, spectral envelope modeling, and Mel-scale feature extraction, which are fundamental in modern speech recognition systems.